# STAT 230A- Final Project Report

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May 2026

## 1 Introduction

Every month, thousands of investors make bets on which stock will perform better. Some win this game, while some lose. But is there a systematic method on how to bet or is it just luck? The Capital Asset Pricing Model, developed in the 1960s offered the first rigorous answer: a stock's return is determined entirely by how much it moves with the market. Yet decades of evidence have shown cracks in this model: when high earnings to price ratio stocks tend to earn higher returns than the CAPM predicts [Basu(1977)] and average returns on small stocks are higher than predicted [Banz(1981)]. This project tests these competing theories using 15 years of monthly returns for the S&P 500 stocks, with an inference-focused goal to understand which factors systematically explain why some stocks earn more than others and whether the celebrated Fama-French model truly accounts for the patterns that CAPM misses.

## 2 Data and Domain Background

### 2.1 The Domain Problem

Stock Markets exist to allocate capital, investors buy shares of companies, and in return expect to be compensated for the risk they take on. The Capital Asset Pricing Model is the one investors turn to make that risk judgement, it says that the only risk that earns a return premium is market risk, which is measured by $\beta$. It reflects the principle that higher expected returns require higher risk [Will Kenton(2026)]. However, there have been quite a few critiques for this model over time. The Fama-French Model introduced in 1993 [Fama and French(1993)] documented two anomalies that CAPM cannot explain: how small cap and value stocks outperform large-cap and growth stocks. Their three-factor model added Small Minus Big(return of small-cap stocks minus large-cap stocks each month) and High Minus Low(return of high book-to-market stocks minus low book-to-market stocks) to market $\beta$, explaining return patterns that had puzzled researchers. This was later on extended further by adding profitability(Robust Minus Weak) and investment(Conservative Minus Aggressive), motivated by the theoretical argument that a firm's value is determined by its expected future profits and how aggressively it reinvests. Separately, [Jegadeesh and Titman(1993)] documented that stocks which performed well over the previous year tends to continue performing well in the near term- the momentum anomaly, which [Carhart(1997)] incorporated into a four-factor model. This project sits squarely in this tradition, using regression analysis to test whether these factors explain the return variation in a modern U.S. equity sample.

### 2.2 The Data

Monthly adjusted closing prices were downloaded for all S&P 500 constituents over the period January 2010 to December 2024 using the Yahoo Finance API via the *yfinance* Python library. Monthly log returns were computed as $r_t = \ln(P_t/P_{t-1})$ giving approximately 180 monthly observations per stock. After removing tickers with insufficient data due to recent IPOs or delistings, the final dataset contains 84,348 stock-month observations across 498 stocks, an unbalanced panel, as not all stocks were listed for the full sample period. The six Fama-French factors:-market excess return (Mkt-RF), size (SMB), value (HML), profitability (RMW), investment (CMA), and momentum (MoM), along with the monthly risk-free rate, were downloaded directly from Kenneth French's Data Library at Dartmouth. The response variable is the monthly excess return for each stock, defined as the stock's log return minus the risk-free rate for that month.

## 3 Final Regression Model

The core idea of the analysis was to work with nested models. A nested sequence is appropriate here because it allows formal incremental F-tests to determine whether each additional factor provides statistically significant explanatory power beyond the previous model, directly addressing the research question. Before defining the final regression model I chose, I introduce the notation used throughout. Let $R_i$ denote the monthly return of stock $i$ and $R_f$ the monthly risk-free rate, so that $R_i - R_f$ is the excess return earned above the risk-free baseline. The intercept $\alpha$ captures unexplained return not attributed to any factor, $\epsilon$ is the stock-specific error term, and $\beta_1, \ldots, \beta_6$ measure each stock's sensitivity to the corresponding factor. The six factors are defined as follows:

**Mkt-RF** Market excess return: captures systematic, market-wide risk.
**SMB** Small Minus Big: return of small-cap minus large-cap stocks, capturing the size premium.
**HML** High Minus Low: return of value minus growth stocks, capturing the value premium.
**MoM** Momentum: return of recent winners minus recent losers, capturing the momentum premium.
**RMW** Robust Minus Weak: return of profitable minus unprofitable firms.
**CMA** Conservative Minus Aggressive: return of low-investment minus high-investment firms.

The four nested models are estimated sequentially, each adding factors to the previous:

**Model 1: CAPM (1 factor)**

$$R_i - R_f = \alpha + \beta_1(\text{Mkt-RF}) + \epsilon$$

**Model 2: Fama-French 3-Factor (3 factors)**

$$R_i - R_f = \alpha + \beta_1(\text{Mkt-RF}) + \beta_2(\text{SMB}) + \beta_3(\text{HML}) + \epsilon$$

Extends CAPM by adding size and value premiums, documented by [Fama and French(1993)].

**Model 3: Fama-French 3-Factor + Momentum (4 factors)**

$$R_i - R_f = \alpha + \beta_1(\text{Mkt-RF}) + \beta_2(\text{SMB}) + \beta_3(\text{HML}) + \beta_4(\text{MoM}) + \epsilon$$

Momentum was not part of the original Fama-French framework; it was documented separately by [Jegadeesh and Titman(1993)].

**Model 4: Fama-French 5-Factor + Momentum (6 factors)**

$$R_i - R_f = \alpha + \beta_1(\text{Mkt-RF}) + \beta_2(\text{SMB}) + \beta_3(\text{HML}) + \beta_4(\text{MoM}) + \beta_5(\text{RMW}) + \beta_6(\text{CMA}) + \epsilon$$

5 factor model from [Fama and French(2015)] and I added momentum along with that in order to test if it provides incremental explanatory power beyond Fama-French 5-factor.

**Model Selection:** Based on the results, the Fama-French 3-Factor model is selected as the final model. The transition from CAPM to FF3 yields the largest and most significant improvement: $\alpha$ shrinks by 68% and becomes insignificant ($p = 0.2429$), with the confidence interval $[-0.000841, +0.000213]$ crossing zero for the first time, meaning zero is a plausible value for $\alpha$. The incremental F-test confirms this ($F = 327.66$, $p \approx 0$), indicating that SMB and HML jointly absorb most of the unexplained return left by CAPM. Adding momentum provides no incremental benefit ($F = 0.63$, $p = 0.426$), and while RMW and CMA are jointly significant, the adjusted $R^2$ gain is negligible (0.2784 to 0.2786) and alpha deteriorates slightly under FF5+MOM ($p = 0.062$). FF3 offers the best balance of parsimony and explanatory power for this sample.

# 4 Discussion

## 4.1 OLS Regression Results

Four OLS models were estimated sequentially, the following table is the result obtained from it:

Table 1: Pooled OLS Regression Results Across Models

| Model | Const | Mkt-RF | SMB | HML | Mom | RMW | CMA | Adj. $R^2$ |
|---|---|---|---|---|---|---|---|---|
| **CAPM** | $-0.0010^{***}$ | $1.0251^{***}$ | – | – | – | – | – | 0.2728 |
| | (0.0003) | (0.0058) | | | | | | |
| **FF3** | $-0.0003$ | $0.9929^{***}$ | $0.1071^{***}$ | $0.1505^{***}$ | – | – | – | 0.2784 |
| | (0.0003) | (0.0061) | (0.0103) | (0.0080) | | | | |
| **FF3_Mom** | $-0.0003$ | $0.9943^{***}$ | $0.1085^{***}$ | $0.1518^{***}$ | 0.0063 | – | – | 0.2784 |
| | (0.0003) | (0.0064) | (0.0104) | (0.0082) | (0.0079) | | | |
| **FF5_Mom** | $-0.0005^{*}$ | $0.9927^{***}$ | $0.1315^{***}$ | $0.1501^{***}$ | 0.0113 | $0.0659^{***}$ | $-0.0127$ | 0.2786 |
| | (0.0003) | (0.0064) | (0.0117) | (0.0113) | (0.0081) | (0.0145) | (0.0168) | |

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Standard errors in parentheses. N = 84,348 for all models.

The regression results support the Fama-French three factor model as the most meaningful upgrade from CAPM. $\alpha$ drops from -0.1% under CAPM to -0.03% under Fama-French-3, indicating that size and value factors collectively absorb the unexplained return. Momentum adds no incremental explanatory power. RMW is significant and positive, while CMA is negative and insignificant, possibly reflecting the outperformance of high-investment technology firms during the sample period. Adjusted $R^2$ remains $\approx 0.28$ across all models, consistent with the well-documented difficulty of explaining individual stock returns with systematic factors alone.

These results are broadly consistent with the Fama-French framework, the systematic shrinkage of $\alpha$ from CAPM to FF3 supports the interpretation that size and value represent genuine risk factors rather than statistical anomalies, while the failure of momentum and the wrong-signed CMA coefficient suggest the framework is incomplete for this sample period.

## 4.2 F-tests

As the next step, I ran F-tests on the model progression to formally test whether each progression was justified.

Table 2: F-Tests for Nested Model Comparisons

| Comparison | Description | F-Statistic | P-Value | Conclusion |
|---|---|---|---|---|
| CAPM → FF3 | Adding SMB and HML | 327.6604 | 0.0 | Significant |
| FF3 → FF3_Mom | Adding Momentum | 0.6336 | 0.426 | Not Significant |
| FF3_Mom → FF5_Mom | Adding RMW and CMA | 10.5378 | 0.0 | Significant |

Significance assessed at the $\alpha = 0.05$ level.

The nested F-tests reveal that Fama-French 3 factor model provides the only substantial improvement over CAPM (F=327.66). Momentum is redundant in this sample, RMW and CMA are jointly significant(F=10.54) despite CMA being individually insignificant, consistent with the HML-CMA correlation observed in the exploratory correlation analysis in Section 5.1.

## 4.3 Alpha Analysis across models

Table 3: Alpha Analysis Across Models

| Model | Alpha | Robust SE | P-Value | Significance | CI Lower | CI Upper | Conclusion |
|---|---|---|---|---|---|---|---|
| CAPM | $-0.000968$ | 0.000274 | 0.0004 | *** | $-0.001506$ | $-0.000431$ | Significant |
| FF3 | $-0.000314$ | 0.000269 | 0.2429 | | $-0.000841$ | 0.000213 | Not significant |
| FF3_Mom | $-0.000343$ | 0.000273 | 0.2094 | | $-0.000878$ | 0.000192 | Not significant |
| FF5_Mom | $-0.000517$ | 0.000277 | 0.0617 | * | $-0.001059$ | $2.5 \times 10^{-5}$ | Not significant |

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. 95% confidence intervals computed using robust standard errors.

Alpha analysis using HC3 robust standard errors confirms that the FF3 model successfully eliminates statistically significant unexplained return, alpha drops from -0.097% (p=0.0004) under CAPM to -0.031% (p=0.243) under FF3, with the confidence interval crossing zero for the first time. Adding momentum leaves alpha unchanged, while the FF5+MOM model shows a slight deterioration to marginal significance (p=0.062), likely attributable to CMA's negative coefficient partially offsetting RMW's positive contribution. These results support FF3 as the most complete specification for this sample. Figure 1 below is a better representation of the sharp upward jump from CAPM to FF3, where the confidence interval crosses zero for the first time. This visually confirms that the Fama-French three-factor model successfully eliminates statistically significant unexplained return.

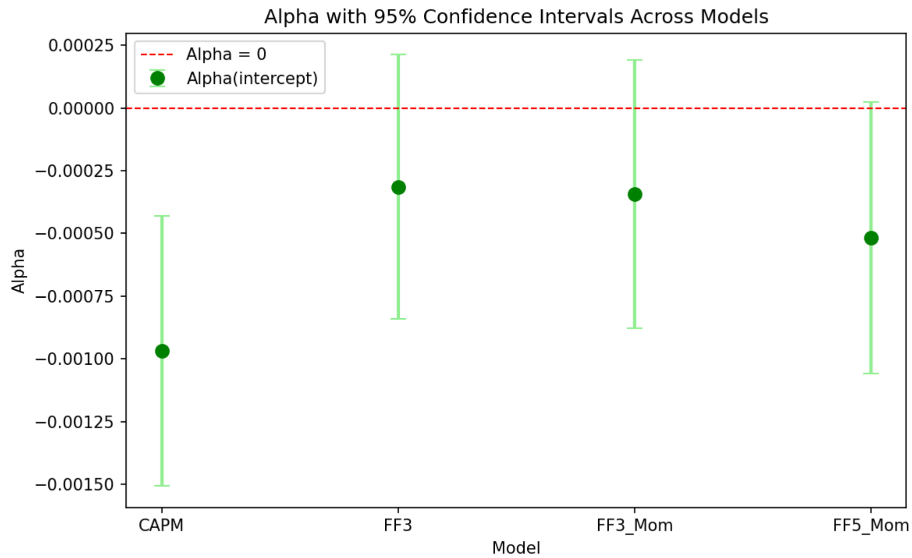


Figure 1: Alpha Analysis

### 4.4 Implications

These findings carry practical implications for different audiences, it could be for people who want to invest their savings or portfolio managers seeking to improve their returns. From the analysis, I can say that investors looking to beat the market should consider favouring smaller companies and cheaper "value" stocks, since these are the only characteristics that reliably predicted higher returns in this sample. Strategies that simply chase recent winners, however, did not show a meaningful edge among large S&P 500 stocks. Finally, one traditional signal, that cautious, low-spending companies outperform aggressive ones, appears to have weakened in recent decades, possibly because today's dominant technology companies operate very differently from the industrial era firms the theory was built on.

### 4.5 Limitations

Several limitations should be considered when interpreting these results: First, restricting the sample to current S&P 500 constituents introduces **survivorship bias**-companies that failed, were acquired, or were delisted during 2010-2024 are excluded, which inflates average returns relative to what a real investor would have earned. Second, the **pooled panel structure** assumes that factor loadings are identical across all 498 stocks, a utility stock and a technology stock have different true sensitivities to SMB and HML, but the model estimates one average coefficient for each factor across the entire cross-section. Third, the **sample period** of 2010-2024 is characterized by a bull market, which may not be representative of long-run factor behavior. In particular, the insignificance of momentum may be due to the specific crash-and-recovery dynamics of 2020 rather than a structural absence of the momentum premium. Fourth, **look-ahead bias** is present in the S&P 500 constituent list: we know today which companies survived, but an investor in 2010 did not. Finally, an alternative approach would be to run firm by firm time-series regressions rather than a pooled panel, which would allow factor loadings to vary across stocks, this is a direction for future work.

# 5 Additional Work

All diagnostics below are run on the most complex model (Fama-French 5 factor + Momentum model) as a conservative check, if assumptions hold for the most complex model, they hold for the simpler Fama-French 3 factor model selected in Section 3.

## 5.1 Exploratory Data Analysis

Prior to fitting models, a correlation matrix (section A) of the six factors was examined. The most notable finding was the moderate correlation between HML and CMA ($r = 0.64$), which anticipated the result that CMA would be individually insignificant despite the joint F-test for RMW and CMA being significant ($F = 10.54$). All other pairwise correlations were below 0.40, suggesting the factors are sufficiently orthogonal for reliable coefficient estimation, a finding later confirmed by VIF values all below 2.26.

## 5.2 Assumption Diagnostics

### 5.2.1 Linearity

From figure 2, we can say that linearity holds since the residuals are centered around zero the entire range of fitted values with no obvious curve. However, it clearly reveals heteroskedasticity.

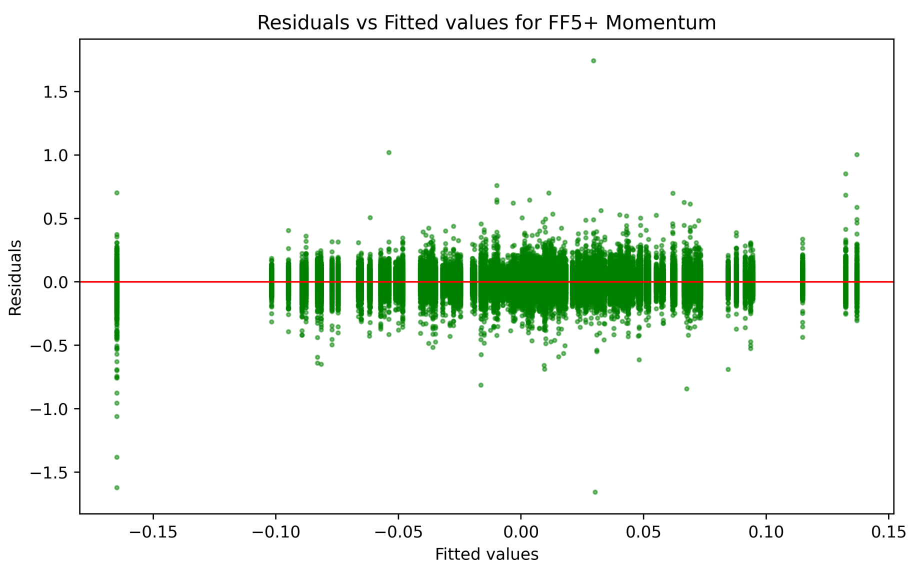


Figure 2: Residual vs Fitted for Fama-French 5 factor+ Momentum

### 5.2.2 Heteroskedasticity

In order to confirm it further, I performed two tests : 1. Breusch Pagan and 2.White Test. Breusch Pagan resulted in a **stat of 339** with $p \approx 0$ which is an overwhelming rejection. The White test which is a more generalised test that also catches nonlinear forms of heteroskedasticity resulted in a **stat of 1738.01** with $p \approx 0$. Thus proving what I concluded from above from Figure 2. Since all the p-values in the regression table 1 were computed with standard OLS standard errors, which are now known to be incorrect, the next step was to perform HC3 robust refitting.

### 5.2.3 Normality

Jarque-Bera formally rejects normality (**stat=1385891**, $p \approx 0$), driven primarily by excess kurtosis of 22.84 — reflecting the fat-tailed nature of equity returns documented extensively in the finance literature. CLT and HC3 robust standard errors mitigate the impact on inference.

The Q-Q plot in Figure 3 reveals fat tails in both directions, which is consistent with well-documented properties of equity returns. While the Jarque-Bera test formally rejects normality, this was expected with such financial data and does not invalidate inference given the large sample size (n=84,348) and the use of HC3 robust standard errors.

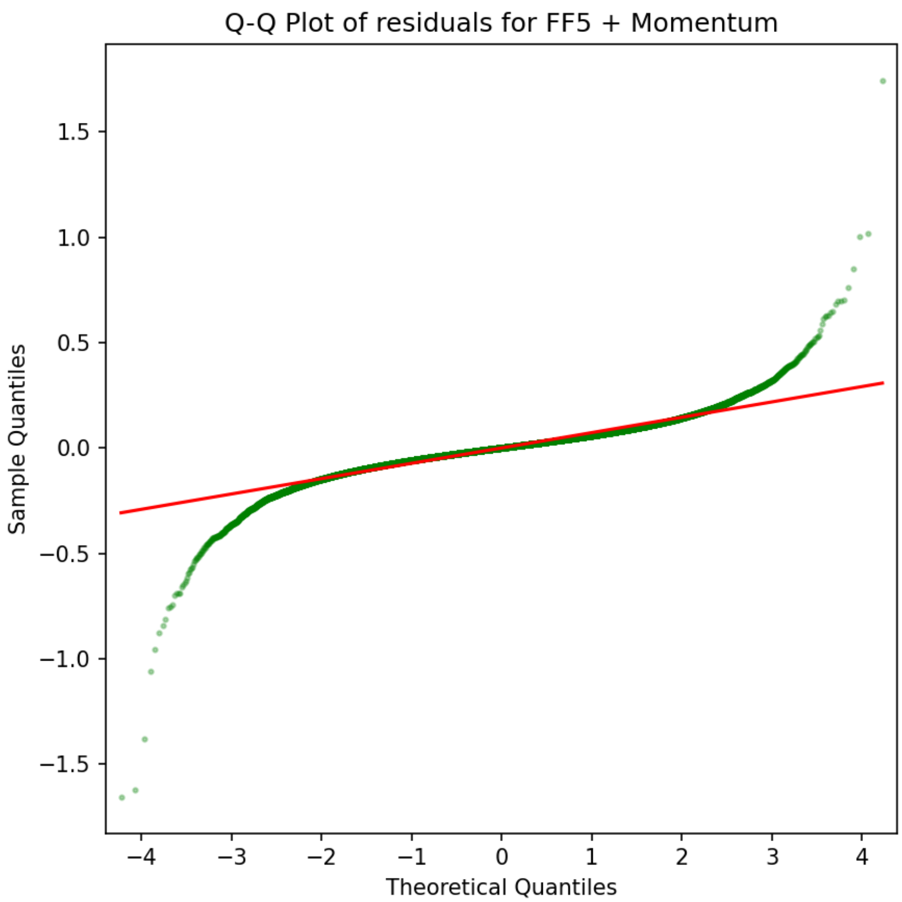


Figure 3: Q-Q plot of residuals

### 5.2.4 Multicollinearity

Table 4: Variance Inflation Factors (FF5_Mom Model)

| **Variable** | **Mkt-RF** | **SMB** | **HML** | **Mom** | **RMW** | **CMA** |
|---|---|---|---|---|---|---|
| **VIF** | 1.2517 | 1.6543 | 2.2530 | 1.3367 | 1.2748 | 1.9660 |

All VIF values well below 10, indicating no severe multicollinearity.

All six values are well below 5, thus multicollinearity is not a concern here.The factors are sufficiently independent that individual coefficient estimates are reliably estimated, and any non-significant results (like MOM and CMA) reflect genuine lack of explanatory power rather than imprecise estimation due to collinearity.

#### 5.2.5 Cook's Distance

Figure 4 reveals a cluster of influential observations around index 55000 consistent with the COVID-19 crash of March-April 2020 where extreme simultaneous stock movements exceeded model predictions. No observation approaches the critical threshold of 1.0, indicating that while crisis-period months are disproportionately influential, they do not individually distort the regression results.

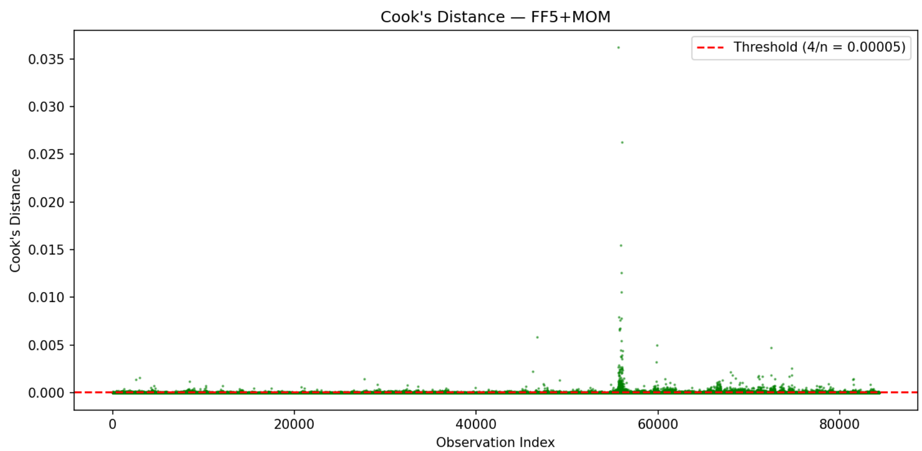


Figure 4: Cook's Distance

## 6 Conclusion

This project tested whether firm-level and market-wide factors explain cross-sectional variation in monthly U.S. equity returns using 84,348 stock-month observations from the S&P 500 over 2010-2024. Four nested OLS models were estimated, progressing from CAPM through to a Fama-French 5-factor model plus momentum. The main finding is that the Fama-French 3-factor model is the most effective model for this sample: the addition of SMB and HML reduces $\alpha$ by 68% and renders it statistically insignificant ($p = 0.243$), with an incremental F-statistic of 327.66, by far the largest improvement in the model progression. Momentum adds no explanatory power ($F = 0.63$, $p = 0.426$), consistent with the known momentum crash during the COVID-19 recovery. RMW and CMA are jointly significant but yield negligible gains in adjusted $R^2$, and CMA's negative coefficient contradicts FF5 theory, likely reflecting the dominance of high-investment technology firms in the sample period. Taken together, these results broadly support the Fama-French framework while highlighting that not all factors perform equally across market regimes.

# 7 Bibliography

## References


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# A Exploratory Data Analysis

I performed a basic EDA on the data before fitting the models, no duplicates and null values were found.

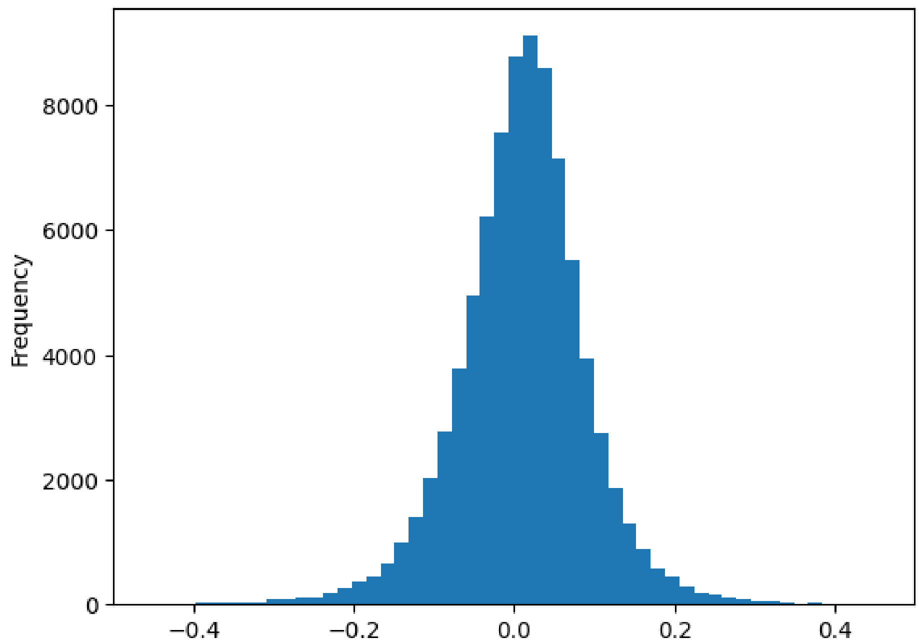


Figure 5: Distribution of Excess Return

Figure 5, the histogram shows us that the 'Excess Returns' is slightly left skewed and has an excess positive kurtosis. It means extreme returns happen more often than a normal distribution would predict.

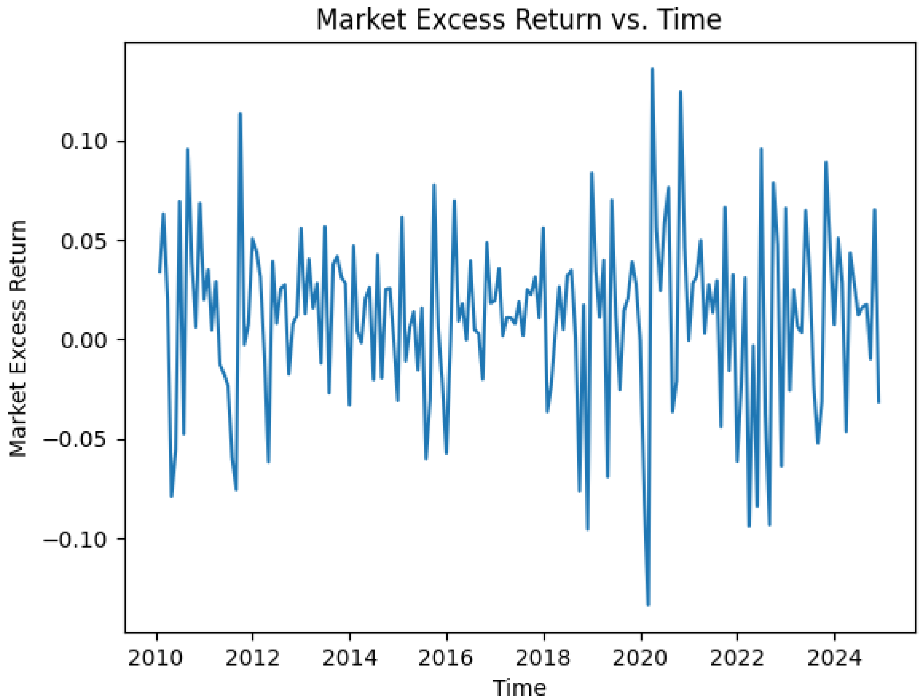


Figure 6: Mkt-RF vs Time

Volatility: As expected, the swings get larger around 2010-11 (European Debt crisis) and 2020(COVID). Calm periods and turbulent periods tend to cluster together. Also, Market Excess Return fluctuates around zero, as expected. This is because it's an excess return and not price.

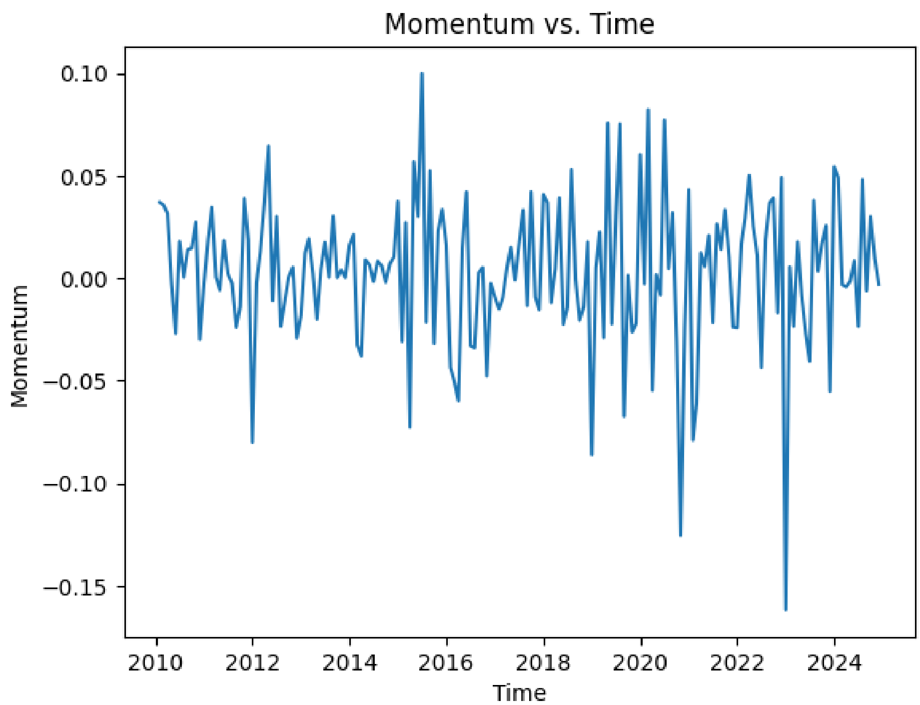


Figure 7: Momentum vs Time

Momentum tends to sharply reverse during market recoveries (like 2009, 2020) because recent losers suddenly outperform.

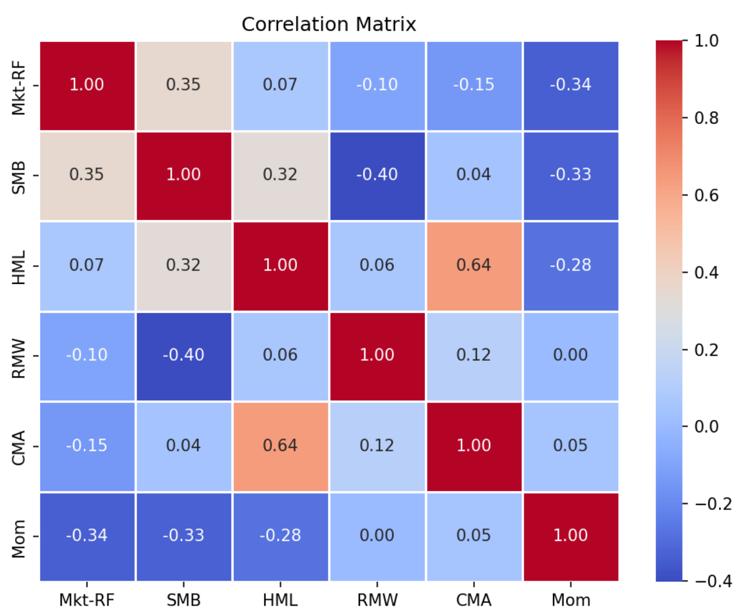


Figure 8: Correlation Matrix from EDA

The correlation matrix reveals that the six factors are largely uncorrelated by construction, supporting their use as independent predictors. The most notable exception is HML and CMA ($r = 0.64$), which reflects the well-documented overlap between value and conservative investment strategies.